1. **Sensor Detection Subsystem**.

Component	Specification	Price (BDT)
mmWave Radar	3X TI IWR6843 / AWR1843 Radar Module	150,000
IR PTZ Camera	Night Vision Tracking Camera	50,000
Acoustic Detection Array	4 MEMS Microphone Array	50,000
LRF	2000m Laser Rangefinder	50,000
RF Detector Module	2400–2500 MHz	100,000
Total:		**400,000**

2. **AI Processing & Control**.

Component	Specification	Price (BDT)
AI Computing Core	NVIDIA Jetson Nano 4GB Module	60,000
Microcontroller	STM32F4 Control Board	50,000
Storage & Interface	SSD, Communication Interface	10,000
Total:		**120,000**

3. **Command & Monitoring Console**.

Component	Specification	Price (BDT)
Display Unit	10" Industrial Display	15000
Operator Keyboard	Rugged Keyboard	2000
Joystick Controller	PTZ Control Joystick	3000
Total:		**20,000**

4. **Power & Communications**.

Component	Specification	Price (BDT)
Power Supply	Custom PSU	10000
UPS Backup	Battery Backup System	10000
Communication Module	Ethernet / Wi-Fi	10000
Total:		**30,000**

5. **Mechanical Structure**.

Component	Specification	Price (BDT)
Weatherproof Enclosure	IP65 Rated	50000
Mechanical Frame	Tower & Mounting Structure	50000
Total:		**100,000**

6. **Software Development & Integration**.

Component	Specification	Price (BDT)
AI Detection Software	YOLO-Based Drone Detection	30,000
Dashboard & Logging	Monitoring Interface	20,000
Integration & Testing	System Calibration	30,000
Total:		**80,000**

Budget Summary

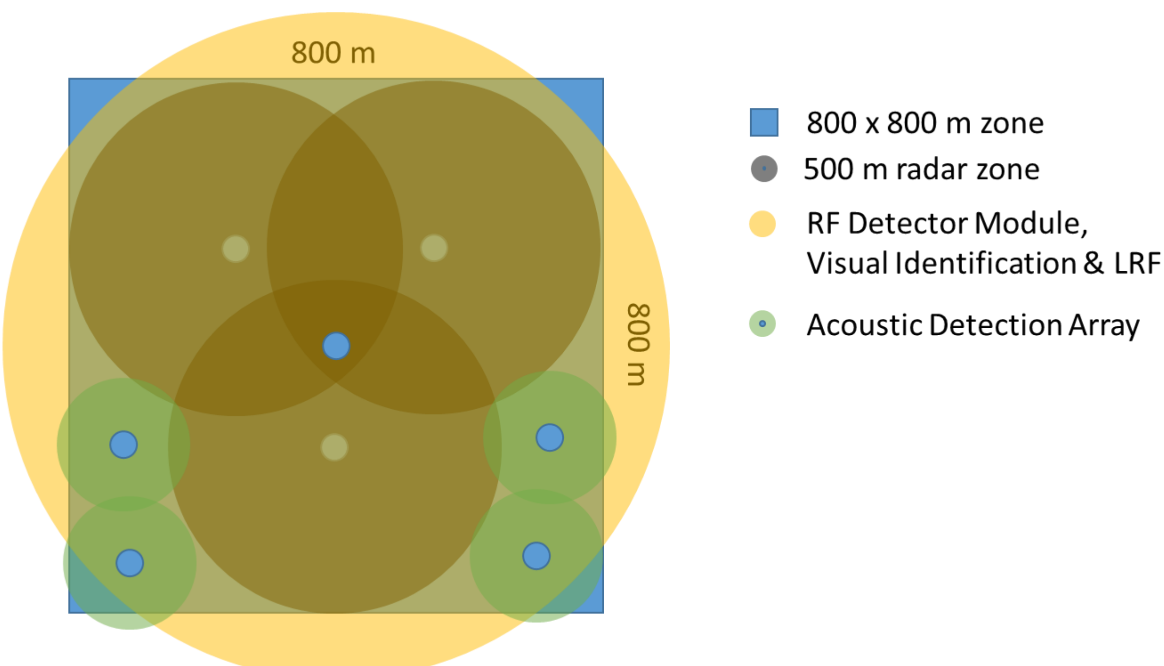
Category	Cost (BDT)
Sensor Detection Subsystem	400,000
AI Processing & Control	120,000
Command & Monitoring Console	20,000
Power & Communications	30,000
Mechanical Structure	100,000
Software Development & Integration	80,000
Total:	**750,000**

Optional Upgradation Plan to C-UAS Soft Kill

Component	Price (BDT)
RF Power Amplifier (Jammer Module) per model	100,000
1 Additional Radar Module	50,000
4xAcoustic Detection Array	50,000
GNSS Spoofer Module	100,000
Total:	**300,000**

Technical Capability

Capability	Expected Performance
Radar Detection	500 m
Visual Identification (Day)	1000m
Visual Identification (Night)	800 m
Acoustic Detection	200 m
RF Detection Range	1000m (passive)
Laser Rangefinder	1000m
Object Classification	High for visible targets
Tracking Accuracy	±10–15°
Multi-Target Tracking	20–30 Targets
Update Rate	10–30 Hz
Operating Hours	24/7
Power Requirement	12–24 VDC || 120–240 VAC



Technical Specification

mmWave Radar Subsystem

Parameter	Specification
Model	TI AWR1843
Radar Type	FMCW mmWave Radar
Frequency Band	76–81 GHz (AWR1843)
Bandwidth	Up to 4 GHz
Source	**https://mou.sr/4fcJNor**
Range Resolution	4–15 cm
Velocity Measurement	Yes
Azimuth Measurement	Yes
Elevation Measurement	Optional
Update Rate	Up to 30 Hz
Detection Range	300–500 m
Multi-Target Tracking	Up to 10 Targets per Radar
Interface	UART | Ethernet
Operating Temperature	-40°C to +85°C
Power Consumption	<10 W

Embedded Controller

Parameter	Specification
MCU Family	STM32F4
CPU Frequency	216 MHz
Flash Memory	Up to 16 MB
SRAM	Up to 4 MB
Communication	UART, CAN, SPI, I2C, USB, Ethernet
Operating Voltage	3.3 V
Real-Time Processing	Yes
Source	**Custom**

Electro-Optical / IR PTZ Camera

Parameter	Specification
Sensor Type	CMOS Sensor
Resolution	1920 × 1080
Frame Rate	30 FPS
Optical Zoom	30×
IR Illumination Range	100–1000 m
Detection Range (Day)	Up to 800 m
Detection Range (Night)	Up to 300 m
Pan Rotation	360° Continuous
Tilt Rotation	±90°
Video Compression	H.264 / H.265
Interface	USB , Ethernet
Power Consumption	<20 W
Source	**https://luminixsky.com/product/skydroid-c20-three-axis-night-vision-gimbal/**

Laser Rangefinder (LRF)

Parameter	**Specification**
Sensor Type	Laser Rangefinder Module
Laser Wavelength	1535 nm (Eye-safe, Class 1)
Measurement Range	50 m – 2000 m
Range Accuracy	± 1 m
Range Resolution	1 m
Measurement Rate	1 – 10 Hz (configurable)
Beam Divergence	≤ 3 mrad
Output Interface	UART (TTL 3.3 V / RS-232)
Baud Rate	9600 / 115200 bps
Operating Voltage	3.3 V – 5.0 V DC
Power Consumption	< 3 W (peak during pulse)
Operating Temperature	-20°C to +60°C
Storage Temperature	-40°C to +85°C
Dimensions (L×W×H)	≈ 68 × 40 × 35 mm
Weight	≈ 90 g
Protection Rating	IP65
Output Data Format	ASCII / Binary (configurable)
Target Reflectivity	≥ 10% (natural targets)
Integration	Compatible with STM32 UART; GPIO trigger supported
Source	**https://www.alibaba.com/product-detail/Long-Range-1535nm-Spectral-UART-1_1601817785192.html**

Acoustic Detection Array

Parameter	Specification
Sensor Type	MEMS Microphone Array
Number of Microphones	8 Elements
Frequency Response	20 Hz – 20 kHz
Acoustic Detection Range	100–200 m
Direction Finding	Yes
Sampling Rate	48 kHz
Interface	USB / UART
Power Consumption	15 W
Source	**Custom**

AI Computing Module

Parameter	Specification
Module	NVIDIA Jetson Orin 4GB
AI Performance	Up to 20 TOPS
CPU	6-Core ARM Cortex-A78AE
GPU	NVIDIA Ampere Architecture
RAM	4 GB LPDDR5
Storage	128 GB SSD
Operating System	Ubuntu Linux
AI Frameworks	TensorFlow, PyTorch, YOLOv8
Power Consumption	7–15 W
Source	**https://store.roboticsbd.com/development-board/1373-nvidia-jetson-nano-developer-kit-b01-4gb-robotics-bangladesh.html**

Command & Control Console

Parameter	Specification
Display	15" LCD Screen
Resolution	1280 × 800
Control Device	TPC156T0001G01V1
Keyboard	Industrial USB
Video Channels	4 Simultaneous
Data Logging	Yes
Alarm Display	Yes
Source	**https://www.dwin-global.com/15-6-inch-g-g-structure-tempered-glass-capacitive-touch-screen-tpc156t0001g01v1-product/**

Communication System

Parameter	Specification
Ethernet	10/100/1000 Mbps
Wi-Fi	IEEE 802.11 b/g/n/ac
GPS	Multi-GNSS
Protocols	TCP/IP, UDP, MQTT
Remote Monitoring	Supported
Source	**Custom**

Power System

Parameter	Specification
Input Voltage	220 VAC
Output Voltage	12 VDC / 24 VDC
Battery Backup	2–4 Hours
UPS Type	Online / Line Interactive
Surge Protection	Yes
Power Consumption	800–1500 W
Source	**Custom**

RF Detection

Parameter	Specification
Parameter	**Specification**
Module Type	RF Logarithmic Detector AD8318
Frequency Range	2400–2500 MHz
Dynamic Range	70 dB
Input Power Range	-65 dBm to +5 dBm
Primary Application	RSSI / RF Power Measurement
Output Signal	Analog DC Voltage (Proportional to dBm)
Active Current	~68 mA (at 5V)
Source	**Custom**

Overall System Performance

Capability	Performance
Drone Detection	100–500 m
Drone Classification	85–95%
Multi-Target Tracking	20–30 Targets
Detection Latency	<10 sec
Tracking Accuracy	±10–15°
Continuous Operation	24/7
Weather Resistance	IP65
System Availability	>95%